

A phase transition in the exactness of the NPA hierarchy at the critical doubly-tilted CHSH functional

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Abstract

Gigena et al. [npj QI **11**, 82 (2025)] proved the exact quantum maximum $c_Q(\alpha, \beta)$ of the doubly-tilted CHSH functional $B_{\alpha\beta} = \alpha\langle A_0 \rangle + \beta\langle B_0 \rangle + \text{CHSH}$ and observed that the NPA hierarchy level required to reach it grows without evident bound toward the critical line $\alpha + \beta = 2$, leaving open “whether any finite level of NPA will be enough...even in the CHSH scenario.” We quantify the mechanism. Writing $s = 2 - \alpha - \beta$ and $\alpha = \beta$: (i) the quantum value leaves the local bound *cubically*, $c_Q = 4 - s + s^3/6 - s^4/36 + O(s^5)$ (derived from the closed polynomial system and matching the known 12-digit value at $s = 0.002$); (ii) each NPA level k overshoots *quadratically*: $c_k(s) = 4 - s + a_k s^2 + O(s^3)$ with $a_{1+AB} = \frac{3}{64}$ (exact, proved below), $a_2 \approx 0.0281$, $a_3 \approx 0.0096$; (iii) consequently “level k is not exact near the critical line” holds whenever $a_k > 0$, and the statement that *the required exact NPA level diverges toward the critical line* (no single finite level suffices as $s \rightarrow 0$) is equivalent to the positivity of the single sequence (a_k) . We prove the supercritical side completely: for *all* $\alpha, \beta \geq 1$ and *every* NPA level $k \geq 1+AB$, the hierarchy is exact, $c_k = 2 + \alpha + \beta = \text{local} = \text{quantum}$ — via three explicit exact rational certificates $Z_0, Z_A, Z_B \succeq 0$ whose nonnegative combination $Z(\alpha, \beta) = Z_0 + (\alpha - 1)Z_A + (\beta - 1)Z_B$ realizes the affine identity $(2 + \alpha + \beta) - B_{\alpha, \beta} \cdot y = \langle Z(\alpha, \beta), \Gamma(y) \rangle$. Thus the hierarchy’s exactness undergoes a *phase transition* at the critical line: exact at every level in the supercritical quadrant (theorem), and exact at *no* level near the critical line in the subcritical region — stated here as Conjecture 2 and proven in the companion paper [14] (see Remark 3). On the subcritical side we certify the first four levels in exact arithmetic: levels $1+AB$, 2, 3 and 4 are each provably not exact (at $s = \frac{1}{10}, \frac{1}{5}, \frac{1}{20}, \frac{1}{40}$ respectively, via rational pseudo-moments beating c_Q , confirmed by Sturm’s theorem), and we compute the almost-quantum overshoot coefficient exactly, $a_{1+AB} = \frac{3}{64}$. We also identify the exact mechanism: rescaling to the critical corner, the degeneracy has a universal form and the limiting obstruction is (a rescaling of) the *Motzkin polynomial*, the classical nonnegative-but-not-sum-of-squares form, so the finite-level failure sits in the restricted-certificate regime rather than in sum-of-squares degree. The phase boundary has a precise geometric reading: by Nie’s finite-convergence theorem (through Marshall’s boundary Hessian condition), a self-tested optimum is finitely NPA-certifiable whenever its boundary Hessian is nondegenerate — contact order exactly two — so the single-tilt functional (a quadratic touch) is finitely exact at level $1+AB$, while the doubly-tilted cubic touch is exactly where this condition fails. Three verified errata in the published polynomial system of Gigena et al. are documented.

*Exact rational certificates and analysis code: <https://github.com/tohafrit/npa-nonexactness>.

1 Setting and known art

$B_{\alpha\beta} = \alpha\langle A_0 \rangle + \beta\langle B_0 \rangle + \sum_{xy} (-1)^{xy} \langle A_x B_y \rangle$, local bound $c_L = 2 + \alpha + \beta$, quantum advantage region $0 < \alpha + \beta < 2$ [1]. We take the symmetric slice $\alpha = \beta = 1 - s/2$, $s \in (0, 2)$. $c_k(s)$ denotes the level- k NPA value [2], computed throughout in the real symmetrized picture (sound and complete for these levels; see the soundness lemma of [3]).

Gigena et al. prove c_Q is the largest root of an explicit sextet of polynomial coefficients in (α, β) and report that at $\alpha = \beta = 0.999$ even NPA level 10 exceeds c_Q by 8.6×10^{-10} [1]. Their published display equations contain three typos, found and verified while reimplementing them (Section 5); the corrected sextic reproduces their quoted $c_Q(0.999, 0.999) = 3.998000001333$ exactly.

2 The cubic law for the quantum value

From the corrected polynomial system, expanding the largest root at $\alpha = \beta = 1 - s/2$:

$$c_Q(s) = 4 - s + \frac{s^3}{6} - \frac{s^4}{36} + O(s^5),$$

verified against high-precision root-finding over three decades of s and against the quoted 12-digit anchor ($s = 0.002$: formula 1.332889×10^{-9} vs exact 1.332891×10^{-9} above c_L). More generally, approaching a point $(2 - \beta, \beta)$ of the critical line with $\beta \in (0, 2)$ fixed, the advantage is cubic with coefficient $\frac{(4-\beta)(2+\beta)}{54\beta(2-\beta)}$, diverging at the corner $\beta \rightarrow 0$ where it crosses over to the quadratic law of the singly-tilted functional. The optimal measurements collapse to compatibility linearly ($1 - \cos \theta_A^* \sim \frac{2+\beta}{3\beta}s$) and the optimal state to product linearly. This flatness — a cubic-touching maximum with linearly degenerating self-tested realization — is the structural reason finite SOS/NPA certificates struggle; Gigena et al. report analytic SOS intractability already at low degree [1].

3 The quadratic overshoot and the reduction

Numerically (solver precision 10^{-8} , values stable under two solvers), each level overshoots the local bound *quadratically*:

$$c_k(s) = 4 - s + a_k s^2 + O(s^3), \quad a_{1+AB} = \frac{3}{64}, \quad a_2 \approx 0.0281, \quad a_3 \approx 0.0096,$$

the ratios $c_k - c_Q$ at successive s -halvings converging to 4 (quadratic), not 8 (cubic). Since $c_Q - (4 - s)$ is cubic:

Remark 1 (reduction, and a precise reading). *For every k : if $a_k > 0$ then $c_k(s) > c_Q(s)$ for all sufficiently small $s > 0$, i.e. level k is not exact near the critical line. We stress the precise statement this yields. Since the exactness thresholds $s_k^{\text{exact}} \downarrow 0$ (each level is exact for $s \geq s_k^{\text{exact}}$; the notation avoids a clash with the arc-interval endpoints s_k^* of [14]), for every fixed $s > 0$ some finite level is exact — the required level $D(s) < \infty$ pointwise. What $a_k > 0 \forall k$ asserts is the uniform failure: the level required to be exact diverges, $D(s) \rightarrow \infty$ as $s \rightarrow 0^+$, so no single finite level is exact throughout a neighborhood of the critical line. This is exactly the open question of [1], and it is a statement about the positivity of one sequence of nonnegative reals. (The strictly stronger claim that some fixed $s > 0$ admits no exact level is not implied and, given the positive $\sim s^3$ margin at fixed s , is a separate, harder conjecture we do not make.)*

Conjecture 2. $a_k > 0$ for every finite k (with $a_k \downarrow 0$, consistent with NPA convergence).

Remark 3 (resolution). Conjecture 2 is now a theorem: in the companion paper [14] we prove that for every finite level k there is an $\varepsilon_k > 0$ with $c_k(s) > c_Q(s)$ on $(0, \varepsilon_k]$, via an explicit level-independent signed witness for the second-order tangent program at a fixed rational base of the critical face, a uniform rational Slater construction, and an exact (no-repair) arc-feasibility lemma; every layer is verified in exact arithmetic, and the chain has been re-verified end to end against an independently written implementation. Quantitative unconditional instances: $a_2 > \frac{1}{39}$, $a_3 > \frac{1}{188}$, $a_4 > \frac{1}{641}$, whose rational tangent certificates are constructed in the present paper (Remark 4 below) while the arc-feasibility step that turns them into bounds on a_k is proven in [14] (alongside the exact $a_{1+AB} = \frac{3}{64}$ proven here). The present paper retains the conjectural framing in the narrative below, as it documents the phase-transition geometry, the exact anchors, and the certified ladder on which the proof is built.

Remark 4 (the certified rational ladder: $v_2 > \frac{1}{39}$, $v_3 > \frac{1}{188}$, $v_4 > \frac{1}{641}$). The three quantitative bounds quoted above are carried by explicit rational feasible points of a pinned second-order tangent program, constructed as follows (ancillary `verify_v{2,3,4}_rational_base.py`; every check in exact rational arithmetic).

The program. At $s = 0$ the optimal face of $\{\Gamma_k(y) \succeq 0, y_1 = 1, B_0 \cdot y = 4\}$ is affine of dimension k , with coordinates $\beta = \langle A_1 \rangle$ and $\delta_j = \langle (B_1 B_0)^{j-1} B_1 \rangle$, $j = 2, \dots, k$; we pin the base to the rational point $(\beta, \delta_2, \delta_3, \dots) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \dots, \frac{1}{2})$. There the moment-matrix kernel is an explicit integer lattice (the one-sided A_0/B_0 -reduction relations plus the dressings of $R_1 = (1 - A_1)(1 - B_1)$), splitting under party swap into integer bases N_s, N_a . The tangent program maximizes $B_0 \cdot y_2 - y_{1, \langle A_0 \rangle}$ over first-order directions y_1 whose symmetric kernel compression is rank one along an integer lifting direction, $N_s^\top M(y_1) N_s = \lambda u u^\top$ with $\lambda \geq 0$ (at $k = 2$: $u_2 = (1, -1, 0, -1, 0)$; u_3 and u_4 extend it), whose antisymmetric compression vanishes, with homogeneous face pins on y_1, y_2 , and with the two swap-block matrices $\begin{bmatrix} M_0 & M(y_1)N \\ * & N^\top M(y_2)N \end{bmatrix} \succeq 0$. Its value v_k is a lower proxy for a_k : pinning the base point and restricting the first-order support only decrease the tangent value, and the passage from a strictly feasible tangent jet to the curve $c_k(s)$ itself — the implication “ $a_k \geq$ the jet’s objective” — is the no-repair arc-feasibility lemma proven in [14], which consumes exactly the strict Schur margins certified here.

The certificates (blend construction). A solver-grade optimal jet is rationalized (denominators 10^{12}) and blended with a rational weight ε_p toward an exact rational Slater point of the compressed program; the pins are homogeneous and the Schur subtrahend is a PSD-valued quadratic in the first-order jet, so the blended Schur complements gain a strictly positive multiple of the Slater block, and exact feasibility is verified by rational leading minors ($\varepsilon_p = 10^{-54}$ at $k = 2$; 10^{-7} at $k = 3, 4$). The certified objectives, evaluated at the blended jets, give

$$v_2(\frac{1}{2}, \frac{1}{4}) > \frac{1}{39}, \quad v_3(\frac{1}{2}, \frac{1}{4}, \frac{1}{2}) > \frac{1}{188}, \quad v_4(\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}) > \frac{1}{641}$$

(solver-grade reference values 0.02609, 0.00534, 0.00157, against the unpinned $a_2 \approx 0.0281$, $a_3 \approx 0.0096$: the pinned program captures most of the gain). The division of labor between the two papers is one-way: the present paper constructs and certifies the jets; [14] proves the framework lemma and cites these certificates.

Remark 5 (geometric reformulation: curvature at the deterministic vertex). At $s = 0$ ($\alpha = \beta = 1$) the local and quantum values coincide ($c_Q(0) = 4$), attained on the classical face $\langle A_0 \rangle = \langle B_0 \rangle = 1$. A Puiseux expansion of the largest root of the sextic gives, in exact arithmetic,

$$c_Q(s) = 4 - s + 0 \cdot s^2 + \frac{1}{6}s^3 - \frac{1}{36}s^4 + O(s^5),$$

so the quantum curve is cubically flat at this vertex — its second-order coefficient vanishes. Every finite level, being a support function of a spectrahedron, is convex with $c_k(s) = 4 - s + a_k s^2 + \dots$ and $a_k \geq 0$; since $c_k \geq c_Q$ always, $a_k \geq 0$ is automatic. Conjecture 2 is thus the statement that each outer spectrahedral relaxation retains strictly positive curvature $c_k''(0) = 2a_k > 0$ at a boundary vertex where the quantum body itself has none.

For the almost-quantum level itself this curvature is pinned to a rational value. The path of maximizers $y(s)$ approaches, as $s \downarrow 0$, the rational point $\langle A_0 \rangle = \langle B_0 \rangle = \langle A_0 B_0 \rangle = 1$, $\langle A_1 \rangle = \langle A_0 B_1 \rangle = \langle A_1 B_0 \rangle = \langle A_0 A_1 \rangle = \langle A_0 A_1 B_0 \rangle = \frac{1}{2}$, remaining moments 0, whose 9×9 moment matrix has rank 3 (spectrum $\{2, \frac{7 \pm \sqrt{13}}{2}, 0^{(6)}\}$) with an explicit integer kernel encoding $A_0 = B_0 = 1$ and $(1 - A_1)(1 - B_1) = 0$. The envelope identity $c'(s) = -\langle A_0 \rangle(s)$ then gives $a_{1+AB} = -\frac{1}{2} \frac{d}{ds} \langle A_0 \rangle|_0$, and a high-precision solve of the (party-swap symmetric) 10-variable program yields $\frac{d}{ds} \langle A_0 \rangle|_0 = -\frac{3}{32}$ to fifteen digits, i.e.

$$a_{1+AB} = \frac{3}{64} = 0.046875$$

(cross-checked to twelve digits by directly fitting $c_{1+AB}(s) - (4 - s)$). The lower bound half of this is proven in exact arithmetic. Identifying the optimizer's first two Taylor coefficients $y_1, y_2 \in \mathbb{Q}(\sqrt{3})$ (e.g. $y_{1,a} = -\frac{3}{32}$, $y_{1,b} = \frac{\sqrt{3}-1}{96}$), one checks, over $\mathbb{Q}(\sqrt{3})$: $B_0 \cdot y_1 = 0$; the first-order tangency $N^\top L(y_1) N \succeq 0$ (spectrum $\{\frac{3}{4}, 0^5\}$); the Bonnans–Shapiro second-order tangent-set condition $N_2^\top (L(y_2) - L(y_1) M_0^\top L(y_1)) N_2 \succeq 0$ (spectrum $\{\frac{7}{64}, 0^4\}$); and $B_0 \cdot y_2 - m \cdot y_1 = \frac{3}{64}$. By the second-order tangent-set characterization a feasible arc $y(s) = y_0 + s y_1 + s^2 y_2 + o(s^2)$ then exists, giving $c_{1+AB}(s) \geq 4 - s + \frac{3}{64} s^2 + o(s^2)$, i.e. $a_{1+AB} \geq \frac{3}{64} > 0$ (ancillary `verify_a1AB.py`). The matching upper bound closes the value. The optimal second-order dual $Z(s) = Z_0 + s Z_1 + s^2 Z_2$ obeys complementarity $Z(s) M(y(s)) = 0$ (to order s^2) together with the class-sum identities $\langle Z(s), P_v \rangle = -(B_s)_v$; this is a linear system over $\mathbb{Q}(\sqrt{3})$, and one verifies symbolically (field split $\mathbb{Q}(\sqrt{3}) \rightarrow \mathbb{Q}$; ancillary `verify_a1AB_upper.py`) that $\langle Z_2, \mathbf{1} \rangle = \frac{3}{64}$ on its entire solution set, with no free-parameter dependence — the dual value's s^2 coefficient is pinned. Here $Z_0 \succeq 0$ has rank 3 (spectrum $\{\frac{3}{2}, 0.6624, 1.8376, 0^6\}$), a mirror of $M(y_0)$, and a tangent-feasible (Z_1, Z_2) exists ($K_Z^\top Z_1 K_Z \succeq 0$ and the dual Schur complement $\succeq 0$). Hence a dual arc gives $c_{1+AB}(s) \leq 4 - s + \frac{3}{64} s^2 + o(s^2)$, so $a_{1+AB} \leq \frac{3}{64}$. Therefore

$$a_{1+AB} = \frac{3}{64},$$

the lower bound in fully exact arithmetic and the upper bound via the exactly-pinned dual value; the fifteen-digit numerics agree. Level 1+AB overshoots with curvature exactly $\frac{3}{64}$ — the first exact NPA overshoot coefficient in this scenario.

Equivalently, writing $s_k^{\text{exact}} = \inf\{s > 0 : \text{level } k \text{ exact}\}$ (the exactness threshold, with level k exact for $s \geq s_k^{\text{exact}}$ and overshooting on $(0, s_k^{\text{exact}})$), the conjecture is $s_k^{\text{exact}} > 0$ for all k ; numerically $s_{1+AB}^{\text{exact}} \approx 0.70$, $s_2^{\text{exact}} \approx 0.37$, $s_3^{\text{exact}} \approx 0.11$ (rank-collapse criterion: the optimizer drops to the rank-4 self-testing moment matrix exactly where the level becomes exact; this flat-extension diagnostic is more robust near the threshold than bisecting the $\sim 10^{-7}$ gap), a decreasing sequence whose positivity is the open content. This is a local differential-geometric obstruction, complementary to the operator-algebraic degree bounds discussed below.

The reliably computable coefficients suggest a sharper form. The leading coefficient a_k is itself the value of a convex program: writing the overshoot optimizer as $h(s) = s g_0 + s^2 g_1 + O(s^3)$, the second-order feasibility $S + s \Gamma(g_0) + s^2 \Gamma(g_1) \succeq 0$ (with S the flat-ray moment matrix) is,

on $\ker S$, a Schur complement — hence *linear* in (g_0, g_1) — and

$$a_k = \max_{g_0, g_1} \left\{ -\langle Z_0, \Gamma(g_1) \rangle - m \cdot g_0 : \langle Z_0, \Gamma(g_0) \rangle = 0, \Gamma(g_0)|_{\ker S} \succeq 0, \begin{bmatrix} \Gamma(g_1)|_{\ker S} & * \\ * & I \end{bmatrix} \succeq 0 \right\},$$

where $Z_0 \succeq 0$ is the critical dual certificate at $s = 0$ ($4 - B_0 y = \langle Z_0, \Gamma(y) \rangle$) and $m = \frac{1}{2}(e_{A_0} + e_{B_0})$. Solved uniformly across levels this gives the lower-bound values 0.048, 0.020, 0.0062 at $k = 1+AB, 2, 3$. The Schur linearization is a restriction of the tangent program, so these are valid *lower* bounds only; the true coefficients are $\frac{3}{64} \approx 0.0469, \approx 0.0281, \approx 0.0096$, with irregular successive ratios (≈ 0.60 , then ≈ 0.34) — the smooth geometric appearance of the proxy values does not persist in the true coefficients. (Level 4, $N = 41$, already sits at the edge of floating-point SDP resolution: its coefficient drifts between 0.0034 and 0.0060 under tolerance tightening and the solver reports the solution inaccurate, so we do not report it; an exact-rational Z_0 or a multiprecision solver would be needed past $k = 3$.) Every reported a_k is strictly positive, and the optimal g_0 concentrates on the top word length (56% at $1+AB$), matching the mechanism by which each level's gain is cut by the new longest-word constraints. Any geometric lower bound $a_{k+1} \geq r' a_k$ with $r' > 0$ would imply Conjecture 2, reducing it to a level-to-level comparison rather than a uniform estimate — a potentially more tractable route, though on the present three reliable levels it remains suggestive.

Individual instances are certifiable at the standard of proof independently of the general theorem of [14], and we do so for the first *four* levels. An exact rational feasible pseudo-moment (ancillary `cert_subcrit_1AB.json`, checked by `verify_subcritical.py`) proves, in exact arithmetic, that level $1+AB$ is *not* exact at $s = \frac{1}{10}$: it is positive semidefinite with $y_1 = 1$, and its objective $\frac{40897847}{10485760} \approx 3.90032$ exceeds the quantum value c_Q — the largest real root of the (corrected) sextic of [1], whose root-freeness above the objective is verified by Sturm's theorem. The same construction one level up (`cert_subcrit_L2.json`, `verify_subcritical_L2.py`) gives an exact level-2 pseudo-moment — positive semidefinite on the full 13×13 level-2 moment matrix — with objective $\approx 3.80155 > c_Q(\frac{1}{5})$, proving level 2 is not exact at $s = \frac{1}{5}$, and once more at level 3 (`cert_subcrit_L3.json`, `verify_subcritical_L3.py`): an exact pseudo-moment PSD on the full 25×25 level-3 moment matrix with objective $\frac{42413102039438377}{10737418240000000} \approx 3.9500279 > c_Q(\frac{1}{20})$, so level 3 is not exact at $s = \frac{1}{20}$; and once more at level 4 (`cert_subcrit_L4.json`, `verify_subcritical_L4.py`), PSD on the 41×41 level-4 moment matrix with objective $\approx 3.9750030 > c_Q(\frac{1}{40})$, so level 4 is not exact at $s = \frac{1}{40}$. The overshoot windows contract sharply with level (thresholds $s_k^{\text{exact}} = 0.70, 0.37, 0.11, \dots$ decaying geometrically, $s_4^{\text{exact}} \approx 0.036$): level 2 is exact for $s \gtrsim 0.37$, level 3 for $s \gtrsim 0.11$, level 4 for $s \gtrsim 0.036$, with peak subcritical gaps $\approx 3 \times 10^{-4}, 10^{-5}, 6 \times 10^{-7}$. These are the subcritical counterparts of the exact supercritical certificates of Theorem 6; the open content of Conjecture 2 is the uniform-in- k statement, not any single level.

This admits a precise placement in recent operator-algebra results. The Bell algebra here is $\mathbb{C}[\mathbb{Z}_2^{*2} \times \mathbb{Z}_2^{*2}]$, and Klep, Levenson and McCullough [5] prove a degree-bounded Fejér–Riesz factorization for exactly this family (they single out the CHSH functional): a trigonometric polynomial that is *uniformly strictly positive* over all unitary representations factors as B^*B with a bounded one-party word length, and *strict positivity is essential* — at margin zero the bounded-degree conclusion can fail [5, 6]. Our operator $c_Q(s)I - \hat{B}_s$ is positive semidefinite but loses strict positivity as $s \rightarrow 0$: its margin is the well depth, of order s^3 . This is precisely the regime their theorem does not control from below. Concretely they show a margin ε *suffices* for some finite degree $M'(\varepsilon)$ but give no estimate for M' ; Conjecture 2 is the complementary

necessity statement — that any certificate at margin $\varepsilon \sim s^3$ requires word length growing as $s \rightarrow 0$. (In the minimal scenario the quantum value is attained [4], so NPA does converge at each fixed s ; the content is the divergence of the required level.) A structural observation sharpens which degree must diverge: the KLM bound controls the *Alice-side* (one-party) word length, leaving the *Bob-side* degree $M'(\varepsilon)$ uncontrolled. The singly-tilted functional $\alpha\langle A_0 \rangle + \text{CHSH}$, which engages only the Alice marginal, is exact already at level $1+AB$ for *all* α — including at its own critical point $\alpha = 2$ (verified: $c_{1+AB} = \sqrt{8+2\alpha^2} = c_Q$ to 10^{-8}) — consistent with a bounded Alice-side certificate. The doubly-tilted functional additionally forces the Bob marginal ($B_0 = 1$ at the vertex, alongside $A_0 = 1$), and it is precisely this Bob-side degree that Conjecture 2 asserts diverges. Probing this with asymmetric moment matrices (Alice word length fixed at 1, Bob length varied), the minimal Bob length yielding exactness appears to increase as $s \rightarrow 0$ ($M' \approx 3$ near $s = 0.1$, larger for smaller s), though the exactness threshold sits at the solver’s precision floor so we do not report sharp values (and by $A \leftrightarrow B$ symmetry a mirror certificate can carry the growing length on Alice instead). Consistently with a diverging degree, we certify in exact arithmetic that the required NPA level exceeds k at a sequence of shrinking tilts: levels $1+AB, 2, 3, 4$ are each not exact at $s = \frac{1}{10}, \frac{1}{5}, \frac{1}{20}, \frac{1}{40}$, and the exactness thresholds s_k^{exact} decay geometrically ($0.70, 0.37, 0.11, \dots$; extrapolating $s_{10}^{\text{exact}} \sim 10^{-4}$, consistent with the level-10 non-tightness at $s = 0.002$ reported in [1]). Establishing the quantitative margin-to-Bob-degree lower bound is the concrete open step, complementary to the dual pseudo-moment route.

4 The supercritical phase is exact at every level

Theorem 6 (supercritical exactness on the quadrant $\alpha, \beta \geq 1$). *There are explicit rational matrices $Z_0, Z_A, Z_B \succeq 0$ (size 9×9 , level $1+AB$; ancillary `cert_Z0_critical.json`, `cert_ZA.json`, `cert_ZB.json`; positive semidefiniteness and all identities verified in exact rational arithmetic by `verify_supercritical.py`) with the affine identities, valid for every normalized moment vector y ,*

$$4 - B_{1,1} \cdot y = \langle Z_0, \Gamma(y) \rangle, \quad 1 - \langle A_0 \rangle = \langle Z_A, \Gamma(y) \rangle, \quad 1 - \langle B_0 \rangle = \langle Z_B, \Gamma(y) \rangle.$$

By linearity, $Z(\alpha, \beta) := Z_0 + (\alpha-1)Z_A + (\beta-1)Z_B$ satisfies $(2+\alpha+\beta) - B_{\alpha,\beta} \cdot y = \langle Z(\alpha, \beta), \Gamma(y) \rangle$. For all $\alpha, \beta \geq 1$ this is a nonnegative combination of positive semidefinite matrices, hence $Z(\alpha, \beta) \succeq 0$, certifying $B_{\alpha,\beta} \cdot y \leq 2 + \alpha + \beta$ at level $1+AB$ and therefore at every level. Thus $c_k(\alpha, \beta) = 2 + \alpha + \beta$, the local (and, by [1], quantum) value, attained by the deterministic point: the NPA hierarchy is exact, at all levels, on the entire quadrant $\{\alpha, \beta \geq 1\}$ (the symmetric ray $\alpha = \beta = 1 + \nu$ is its diagonal, with $Z(1+\nu, 1+\nu) = Z_0 + \nu(Z_A + Z_B)$).

The proof is a nonnegative combination — no boundary or convexity argument is needed — so the region is exactly where the coefficients stay nonnegative. (Exactness in fact extends beyond this quadrant, to a curved spectrahedral region and, numerically, to the axes; we state the quadrant result that the three certificates prove outright.) The kernel of $Z_A + Z_B$ is the span of the four flag-class indicator vectors (basis monomials classified by which of A_1, B_1 they contain); its five-dimensional core has margin $1/6$.

Lemma 7. *(Contained in Theorem 6, stated for emphasis.) $c_k(0) = 4$ for every level $k \geq 1+AB$, attained by the deterministic point.*

The kernel of Z_0 is four-dimensional and rational: it contains the all-ones vector, the columns of the “letter-count” matrix $K_{pq} = n_1(w(p, q))$, and one further vector $v_4 = (2, -4, -1, -4, -1, 2, 5, 5, -4)$ forced by complementarity with the second-order optimal face. In particular Z_0 annihilates the entire *flat ray* $y_w(\tau) = 1 - \tau n_1(w)$, on which $B_0 \cdot y \equiv 4$: the critical optimum is not a point but a segment, and the quadratic overshoot of Section 3 emanates from second-order perturbations around interior ray points ($\tau^* \approx 0.51$ at level $1+AB$), whose gain directions live in the kernel of $\Gamma(y(\tau)) = J - \tau K$. At level $1+AB$ this kernel is described exactly by a $\mathbb{Z}_2 \times \mathbb{Z}_2$ flag-class structure (reduced eigenvalues $4 - 4\tau, 2\tau, 2\tau, 0$). The ray remains feasible and optimal through level 2; at level 3 the optimal base curves away from it, and at every level the gain concentrates on the top-degree words — the corner that level- $(k+1)$ consistency constraints do not reach. A proof of Conjecture 2 amounts to a feasible trajectory with positive marginal gain at each level; the two-scale structure of the second-order analysis (a buffer mode orthogonal to $\text{supp } Z_0$, which makes naive second-order formulas underestimate the cost by $\sim 20\%$) is documented in the accompanying analysis records.

5 Errata in the published polynomial system

Reimplementing Ref. [1]: (i) the sextic coefficient printed as $\tau_5 = +4\alpha\beta$ must be $-4\alpha\beta$ (with the printed sign the largest root is far from the quantum value at every tested point, including the symmetric case covered by their own Theorem 1; with the corrected sign it matches direct Bell-operator eigenvalue maximization to machine precision at seven test points, and their own quoted 12-digit benchmark); (ii) the stationarity display equations have α and β interchanged (residuals $O(0.1)$ at the true optimum as printed, 10^{-7} after the swap); (iii) the optimal-cosine formula contains the same swap and a typographic defect in the surd. Their characteristic polynomial and all quoted numerics are correct; the notebooks presumably contain the right expressions. We record these as verified errata so that users of the printed formulas are not misled.

6 The scaling limit is the Motzkin polynomial

The subcritical mechanism has an exact universal form. In the optimal two-qubit representation (Jordan’s lemma), write $A_1 = \cos \varphi_A \sigma_z + \sin \varphi_A \sigma_x$ and likewise B_1 with φ_B ; the Bell operator $\hat{B}_s(\varphi_A, \varphi_B)$ is a degree- $(1, 1)$ trigonometric 4×4 matrix and $c_Q(s) = \max_{\varphi} \lambda_{\max} \hat{B}_s$. Rescale to the critical corner by $\varphi_A = \sqrt{s} x$, $\varphi_B = \sqrt{s} y$.

Theorem 8 (scaling limit). *For fixed (x, y) ,*

$$\lim_{s \rightarrow 0^+} \frac{\lambda_{\max} \hat{B}_s(\sqrt{s} x, \sqrt{s} y) - (4 - s)}{s^3} = U(x, y) := \frac{x^2 y^2 (6 - x^2 - y^2)}{48}.$$

U vanishes on the cross $\{x = 0\} \cup \{y = 0\}$, and $\max_{x,y} U = U(\sqrt{2}, \sqrt{2}) = \frac{1}{6}$, matching the coefficient of s^3 in $c_Q(s) = 4 - s + \frac{s^3}{6} + O(s^4)$.

The proof is a Puiseux expansion of the largest root of the characteristic polynomial of $\hat{B}_s$ (an exact symbolic computation; the value $\max U = \frac{1}{6}$ is an independent cross-check against the quantum expansion of Section 2). The content of the theorem is what U is:

$$\frac{1}{6} - U(x, y) = \frac{1}{6} M\left(\frac{x}{\sqrt{2}}, \frac{y}{\sqrt{2}}\right), \quad M(u, v) = u^4 v^2 + u^2 v^4 - 3u^2 v^2 + 1,$$

where M is the *Motzkin polynomial* — the classical example of a form that is nonnegative but not a sum of squares. Equivalently $P(x, y) := 48(\frac{1}{6} - U) = x^4y^2 + x^2y^4 - 6x^2y^2 + 8 = 8M(x/\sqrt{2}, y/\sqrt{2})$ is nonnegative, is *not* a sum of squares (the degree-3 Gram SDP is infeasible), while $P \cdot (x^2 + y^2)$ is a sum of squares. The degenerate maximizer of U at $(\sqrt{2}, \sqrt{2})$ is the image of the quantum optimum.

Remark 9 (scope of the obstruction). *Theorem 8 exhibits the exact shape of the critical degeneracy as a rescaled Motzkin form. We caution that this does not by itself prove Conjecture 2: the Motzkin zero at the corner is nondegenerate (Hessian eigenvalues 4, 12), and on the positive orthant the scalar Motzkin form is certified at low degree (weighted AM–GM, $u^4v^2 + u^2v^4 + 1 \geq 3u^2v^2$), with no degree growth on a growing box. The finite-level failure is therefore not a scalar-polynomial degree phenomenon; it is a property of the restricted certificate class specific to the hierarchy. We stress the limits of a primal argument: for genuinely operator-valued (Fejér–Riesz) certificates on $\mathbb{T}^2$ there is no degree blow-up in two variables [7], so the finite-level failure of NPA is a property of the restricted certificate class (multipliers of bounded word length in the group algebra $\mathbb{C}[\mathbb{Z}_2^{*2} \otimes \mathbb{Z}_2^{*2}]$), placing it in the dimension-two non-stability regime [8]. Accordingly the natural route to a proof of Conjecture 2 is dual: exhibit, for each k , a degree- k pseudo-moment functional that is positive but admits no flat (Curto–Fialkow) extension, with mass concentrating near the corner in the scaling limit above.*

Remark 10 (abelianization: why the commutative reduction does not suffice). *There is a sharper primal reduction. Because the maximizer is attained and self-tested, let π_θ be the unique optimal two-qubit representation, $\theta = (\theta_A, \theta_B)$ its Bloch angles, and $|\psi_\theta\rangle$ its state. A level- k NPA certificate writes $c_Q(s) - \hat{B}_s = \sum_\lambda L_\lambda^* L_\lambda + (\text{ideal})$ with $\deg L_\lambda \leq k$. Evaluating in π_θ and taking $\langle \psi_\theta | \cdot | \psi_\theta \rangle$ annihilates the ideal terms (they vanish in a true representation) and leaves*

$$p_s(\theta_A, \theta_B) := c_Q(s) - \langle B_s \rangle_\theta = \sum_\lambda \|L_\lambda |\psi_\theta\rangle\|^2 \geq 0,$$

a commutative bivariate trigonometric sum of squares of degree $\leq k$. Since evaluation is a degree-non-increasing $$ -homomorphism, $k \geq D_{\text{comm}}(s)$, the minimal trigonometric SOS-factor degree of p_s on the torus $\mathbb{T}^2$. The reduction is valid, but we record that it does not prove Conjecture 2: the commutative degree $D_{\text{comm}}(s)$ stays bounded. Indeed the Motzkin obstruction is an artifact of the point at infinity (the homogenization vanishes at $[1:0:0], [0:1:0]$), whereas on any fixed compact domain — and $\mathbb{T}^2$ is compact — the degenerating family is a sum of squares at base degree, uniformly in the margin. Concretely, on the box $[-1, 1]^2$ (whose corners are the Motzkin zeros) one has the exact preordering identity*

$$M = (1 - x^2)^2 + x^2(1 - y^2)^2 + x^2(1 - x^2)(1 - y^2),$$

each summand ≥ 0 on the box, and $M + \varepsilon$ merely adds the square ε ; at the finite zero the Hessian is positive definite (4, 12), so M is locally SOS there (Scheiderer’s dimension-two local–global principle then makes it globally so on the compact domain [9]). Numerically the explicit p_s built from the self-tested representation reaches c_Q to 10^{-15} yet has $D_{\text{comm}}(s) = 1$ for s down to 0.02, while the true level is $\gtrsim 4$ there. Consequently the uniform conjecture cannot be reduced to a commutative Positivstellensatz degree bound: the growth is carried by the noncommutative moment structure — the commutator terms $[A_x, B_y]$ that π_θ annihilates — which the abelianization discards. The genuine two-variable-ness still distinguishes double- from single-tilt (the latter is univariate degree-preserving Fejér–Riesz, matching exactness of level $1+AB$), but the missing step lies in the noncommutative hierarchy, not in commutative real algebraic geometry.

7 The contact-order dichotomy

The overshoot is one side of a dichotomy fixing exactly when a tilted CHSH optimum is finitely certifiable; its provable side specializes Nie’s finite-convergence theorem.

Proposition 11 (nondegenerate self-tests are finitely exact). *Let f be a Bell functional whose quantum optimum p^* is attained and self-tested, and pass to the commutative polynomial optimization over the qubit measurement angles supplied by Jordan’s lemma. If the Boundary Hessian Condition holds at the optimizer — equivalently constraint qualification, strict complementarity and second-order sufficiency all hold, i.e. the boundary Hessian of the quantum value along f is nondegenerate (contact order exactly two) — then the NPA hierarchy is exact at a finite level.*

This is Nie’s theorem [10], whose proof runs through Marshall’s Boundary Hessian Condition [11]; the noncommutative statement follows either by the Jordan reduction to a compact commutative program or directly through flat extension. The condition is generic (it holds on a Zariski-open set), so finite exactness is the generic situation and the overshoot is a codimension-one, tuned phenomenon.

Corollary 12 (single tilt). *The single-tilt functional $\alpha\langle A_0 \rangle + \text{CHSH}$, $\alpha \in [0, 2)$, is finitely NPA-exact at level $1+AB$, with $c_Q(\alpha) = \sqrt{8 + 2\alpha^2}$.*

Here $c_Q(\alpha) = \sqrt{8 + 2\alpha^2}$ is smooth with $c_Q'' \neq 0$ (e.g. $c_Q''(2) = \frac{1}{4}$): a nondegenerate quadratic touch, so Proposition 11 applies; an explicit certificate is Bamps–Pironio’s [12].

The doubly-tilted family is precisely the *failure* locus of Proposition 11. Along the critical line the boundary Hessian degenerates — the cubic touch $c_Q(s) = 4 - s + 0 \cdot s^2 + \frac{1}{6}s^3 + \dots$ (contact order three; the value responds as $\delta^{3/2}$ rather than δ^2), so strict complementarity fails and the Condition is silent. Whether finite exactness then *fails* — Conjecture 2, $a_k > 0$ for all k — is the converse, proven in [14]; deciding finite convergence is NP-hard in general [13], so no soft converse was expected, and indeed the proof produces a positive certificate at the symmetric level itself (a signed witness in the non-quantum tangent cone).

Remark 13 (the planar picture; how a finite level rounds the cusp). *The dichotomy is planar: $c_k(s)$ is the support function of the convex body $P_k = \{(\langle A_0 \rangle + \langle B_0 \rangle, \text{CHSH})\} \subset \mathbb{R}^2$, and in the coordinates $t = 2 - u$, $e = 2 + t - w$ the quantum boundary at the shared deterministic vertex is the exact semicubical cusp $9e^2 = t^3$. The Puiseux type of a level’s boundary is controlled by its overshoot through the Legendre map: $c_k - (4 - s) \sim a_k s^2$ with $a_k > 0$ makes the map a local diffeomorphism, giving an unramified analytic branch of curvature $e''(0) = 1/(8a_k)$, whereas the quantum $s^3/6$ makes it fold, giving the ramified $e = \frac{1}{3}t^{3/2}$; the parabolic cap has scale $t_k^* = (16a_k/3)^2 \rightarrow 0$, so the cusp emerges only in the $k \rightarrow \infty$ limit, discontinuously in the Puiseux type. For the almost-quantum level this rounding curve is exact and lives in $\mathbb{Q}(\sqrt{3})$:*

$$e_1(t) = \frac{4}{3}t^2 + \frac{-200+8\sqrt{3}}{27}t^3 + \frac{15476-1336\sqrt{3}}{243}t^4 + \dots,$$

equivalently $c_{1+AB}(s) = 4 - s + \frac{3}{64}s^2 + \frac{25-\sqrt{3}}{512}s^3 + \frac{1783-116\sqrt{3}}{49152}s^4 + \dots$ (machine-verified, ancillary). Two structural contrasts then quantify the obstruction. First, the quantum branch data at the vertex is rational and ramified while every computed finite-level branch is unramified with coefficients in a nontrivial extension, and the extension explodes with the level: $a_{1+AB} = 3/64 \in \mathbb{Q}$, the level- $(1+AB)$ jet lies in $\mathbb{Q}(\sqrt{3})$, while $a_2 = 0.02805867220535\dots$ (certified to 480 digits)

admits no minimal polynomial of degree ≤ 32 with coefficients $\leq 10^8$ (nor degree ≤ 16 up to 10^{18}) — so no proof of Conjecture 2 can proceed through closed-form level values. Second, by Scheiderer’s theorem every convex semialgebraic subset of the plane is a spectrahedral shadow, and the cusp body itself is one (the $(2, 2, 2)$ quantum set is semialgebraic via Jordan’s lemma); hence the conjecture asserts that the NPA family in particular — not finite SDP representations as such — never resolves the cusp, and any proof must use the moment-matrix structure beyond generic spectrahedral-shadow geometry.

8 Outlook

Conjecture 2 is proven in the companion paper [14] (Remark 3); the present paper supplies the geometry the proof is built on: the exact anchor (Lemma 7), the certified strict separations at low levels [3], the exact cubic law of the quantum side, the phase-transition picture, and the contact-order dichotomy (Section 7). Two diagnostic facts recorded here retroactively explain the shape the proof had to take. First, the obstruction is genuinely noncommutative (Remark 10): the commutative shadow certifies at bounded degree, so the growth lives in the uncontrolled Bob-side degree of the noncommutative Fejér–Riesz factorization — and indeed the proof’s witness is a *signed* functional that no Hilbert-space state and no smooth curve of quantum models can realize (in a Krein space the argument of [14] forces the gain direction to be neutral, and no claim is made). Second, the asymmetric Alice-degree-one truncations, natural to the Fejér–Riesz picture, are the wrong object ($\text{Asym}(1, m)$ is a sub-basis of the symmetric level $m+1$, so its overshoot only *upper*-bounds a_{m+1}); the proof accordingly works with a positive certificate at the symmetric level itself, against the right-hand curvature at the convex kink ($c_k''(0^-) = 0$, $c_k''(0^+) = 2a_k$) that no convexity or one-sided-Taylor argument alone could force positive. What remains open after [14] is quantitative, not qualitative: the decay law of the sequence $a_k \downarrow 0$ (the proof gives positivity with witness-dependent constants, not rates), and the sharp dimension law of the first-order space.

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